1. 1) State and prove the *minimum modulus principle* for non-vanishing holomorphic functions f on a domain Ω.
2) Let Ω be a bounded domain and $f \in A(\Omega) \cap C^0(\overline{\Omega})$. Suppose that $|f(z)|$ is constant for $z \in \partial\Omega$. Show that either f is constant or f has a zero in Ω.

2. Prove the following *Painlevé's Theorem*: Assume that Ω_1 and Ω_2 are two domains with an adjacent piece-wise smooth boundary arc L. Let $f_j(z)$, $j = 1, 2$, be holomorphic functions on Ω_j which are continuous on $\overline{\Omega}_j$, respectively. Furthermore, suppose that $f_1(z) = f_2(z)$ for $z \in L$. Define

$$F(z) = \begin{cases} f_1(z), & z \in \Omega_1 \cup L, \\ f_2(z), & z \in \Omega_2. \end{cases}$$

Then, F is holomorphic on $\Omega_1 \cup L \cup \Omega_2$.

3. Let Ω be a domain. A *chain* on Ω is of the form $\sum_{j=1}^m c_j\gamma_j$, where $c_j \in \mathbb{Z}$ and γ_j's are closed piece-wise smooth curves on Ω. A chain $\sum_{j=1}^m c_j\gamma_j$ is said to be *homologous to 0*, if for all $z \in \mathbb{C}\backslash\Omega$,

$$\sum_{j=1}^m c_j W(\gamma_j; z) = 0.$$

Show that for $f \in A(\Omega)$, if $\sum_{j=1}^m c_j\gamma_j$ is homologous to 0 then

$$\sum_{j=1}^m c_j \oint_{\gamma_j} f(z)dz = 0,$$

and

$$\frac{1}{2\pi i}\sum_{j=1}^m c_j \oint_{\gamma_j} \frac{f(\zeta)}{\zeta - z}d\zeta = f(z)\sum_{j=1}^m c_j W(\gamma_j; z), \quad z \in \Omega\backslash \cup_{j=1}^m \Gamma_j.$$

4. Let Ω be a domain and $f \in A(\Omega)$. Let γ be a closed piece-wise smooth curve in Ω which is homologous to 0. Let $\sigma = f \circ \gamma$ and suppose that σ does not pass through α. Let $\{a_j\}_{j=1}^m$ be the preimages of α under f, with multiplicity r_j, $1 \le j \le m$. Prove that

$$W(\sigma; \alpha) = \sum_{j=1}^m r_j W(\gamma; a_j).$$

5. Let γ be a closed piece-wise smooth curve in $\mathbb{C}$ and denote by Γ its trace. Prove that, for any $n \ge 2$,

$$\oint_\gamma \frac{d\zeta}{(\zeta - z)^n} = 0, \quad z \notin \Gamma.$$